| DS | **Session** | Queue | **Question Information** | Level 1 Challenge 56 |
|---|---|---|---|---|

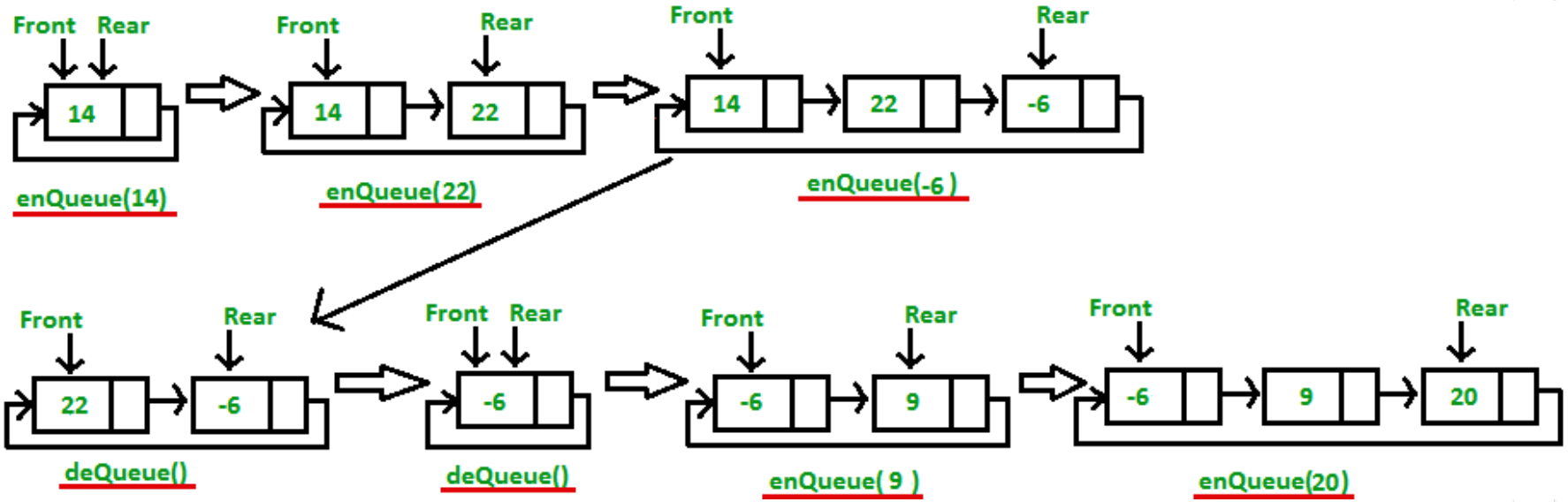
**Question description**

Lalitha is a B.Tech student. During her final year Campus Interview, she has an opportunity to get a job in a software company in Bangalore.

The company provides Five months training period with Rs.30000/month Package. Then it will be incremented to Rs.55000 per month.

At the end of the training, the examination was conducted for all freshers, Lalitha got a question paper and one of the questions comes under the concept of Queues in data structure that is Circular Linked List Implementation in Circular Queue.

Can you help?




**Constraints**

First 3 elements inserted in array0, next 3 elements inserted into array1, remaining elements inserted into array2.

**Input Format**

First line indicates the number of elements to be inserted in the queues.

Second line indicates the elements.

**Output Format**

first line indicates the dequeue element of Q2

Second line indicates the dequeue element of Q1

Third line line indicates the dequeue element of Q0

**Note:** Refer sample input and output test cases

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
10
1 2 3 4 5 6 7 8 9 10
```

EXPECTED OUTPUT

```
Elements in Circular Queue are:1 2 3 4 5 6 7 8 9 10
Deleted value = 1
Deleted value = 2Elements in Circular Queue are:3 4
5 6 7 8 9 10
```

Test Case 2

INPUT (STDIN)

```
10
16 17 18 19 20 11 12 13 14 15
```

EXPECTED OUTPUT

```
Elements in Circular Queue are:16 17 18 19 20 11 12
13 14 15
Deleted value = 16
Deleted value = 17Elements in Circular Queue are:18
19 20 11 12 13 14 15
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void enQueue(Queue* q,int value)
```

Test Case 2

KEYWORD

```
int deQueue(Queue* q)
```

Test Case 3

KEYWORD

```
void displayQueue(struct Queue*
q)
```

Complexity Test Cases

| Test Case 1 | Test Case 2 | Test Case 3 |
| --- | --- | --- |
| CYCLOMATIC COMPLEXITY | TOKEN COUNT | NLOC |
| 2 | 410 | 74 |

# Code

solution.c

```c
#include <stdio.h>

#define MAX 20

typedef struct Queue {
    int arr[MAX];
    int front, rear;
} Queue;

void initQueue(Queue* q) { q->front = q->rear = -1; }

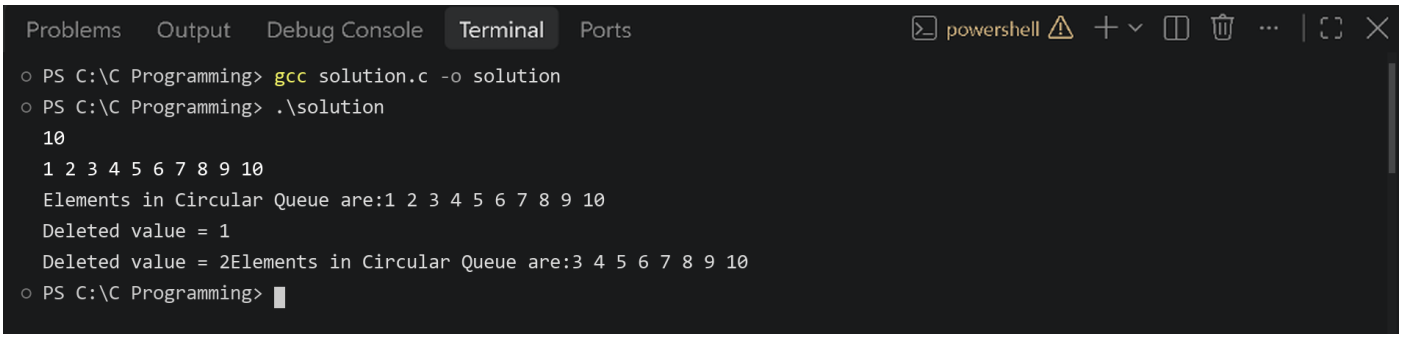
void enQueue(Queue* q, int value) {
    if ((q->front == 0 && q->rear == MAX - 1) || (q->rear == (q->front - 1) % (MAX - 1))) return;
    if (q->front == -1) {
        q->front = q->rear = 0;
        q->arr[q->rear] = value;
    } else if (q->rear == MAX - 1 && q->front != 0) {
        q->rear = 0;
        q->arr[q->rear] = value;
    } else {
        q->arr[++q->rear] = value;
    }
}

int deQueue(Queue* q) {
    if (q->front == -1) return -1;
    int data = q->arr[q->front];
    if (q->front == q->rear) {
        q->front = q->rear = -1;
    } else if (q->front == MAX - 1) {
        q->front = 0;
    } else {
        q->front++;
    }
    return data;
}

void displayQueue(struct Queue* q) {
    if (q->front == -1) return;
    printf("Elements in Circular Queue are:");
    if (q->rear >= q->front) {
        for (int i = q->front; i <= q->rear; i++) printf("%d ", q->arr[i]);
    } else {
        for (int i = q->front; i < MAX; i++) printf("%d ", q->arr[i]);
        for (int i = 0; i <= q->rear; i++) printf("%d ", q->arr[i]);
    }
    printf("\n");
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    Queue q;
    initQueue(&q);
    for (int i = 0; i < n; i++) {
        int x; scanf("%d", &x);
        enQueue(&q, x);
    }
    displayQueue(&q);
    printf("Deleted value = %d\n", deQueue(&q));
    printf("Deleted value = %d", deQueue(&q));
    displayQueue(&q);
    return 0;
}
```

# Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
10
1 2 3 4 5 6 7 8 9 10
Elements in Circular Queue are:1 2 3 4 5 6 7 8 9 10
Deleted value = 1
Deleted value = 2Elements in Circular Queue are:3 4 5 6 7 8 9 10
PS C:\C Programming>